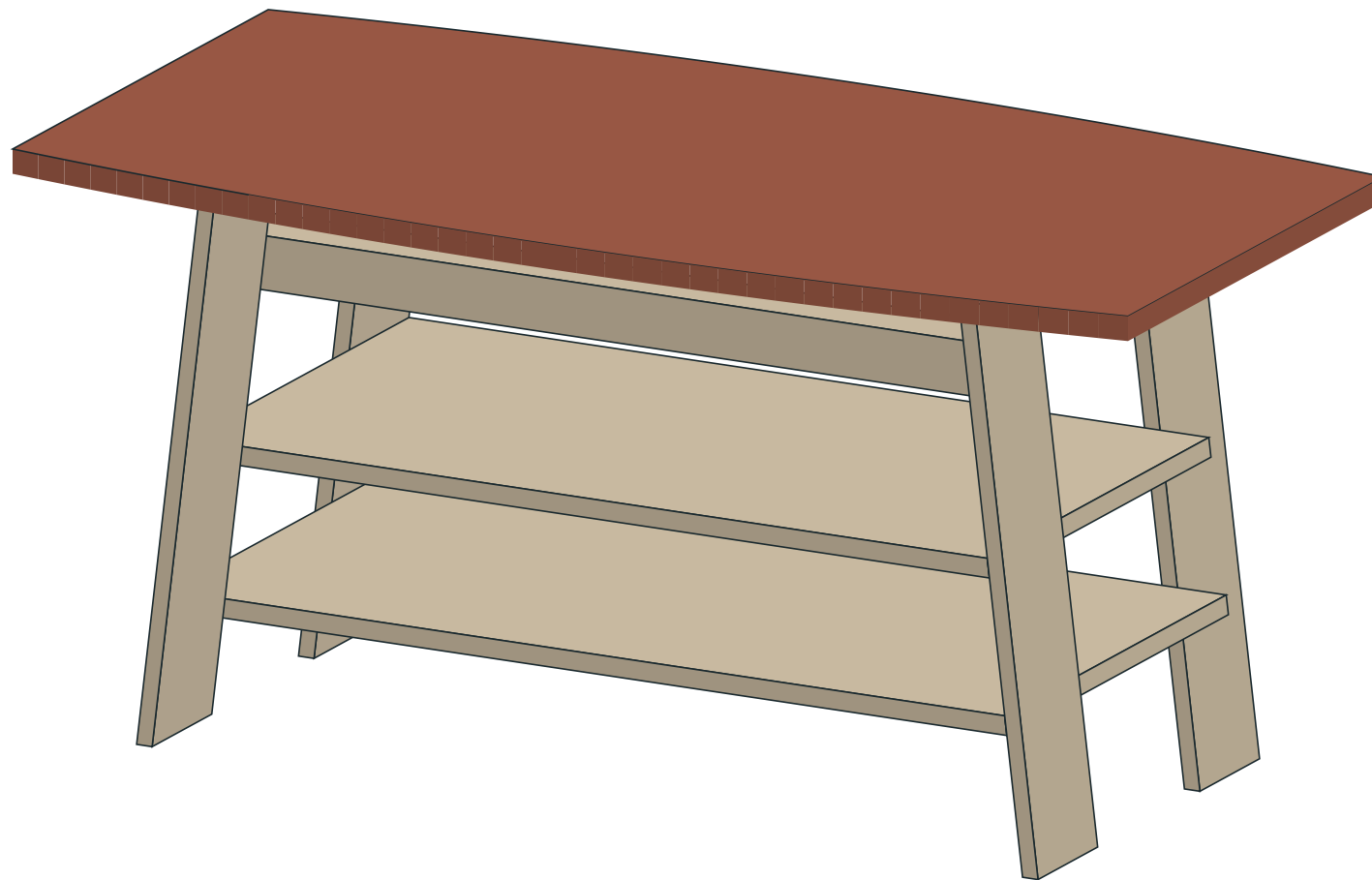


3 / Angled showcase table

A distinct showcase: sloping mating faces, paired shelves, a four-leg structure and coordinated tabletop locking.

FINISHED ASSEMBLY



9 PARTS / 32 LOCKS / 16 MOVEMENTS

1800 W x 760 H x 700 D mm. Four legs splay outward by 8 degrees. Both shelves and both aprons have ends cut to the same angle. The muted terracotta tabletop highlights the final four-leg connection.

Exploded view / angled mating faces

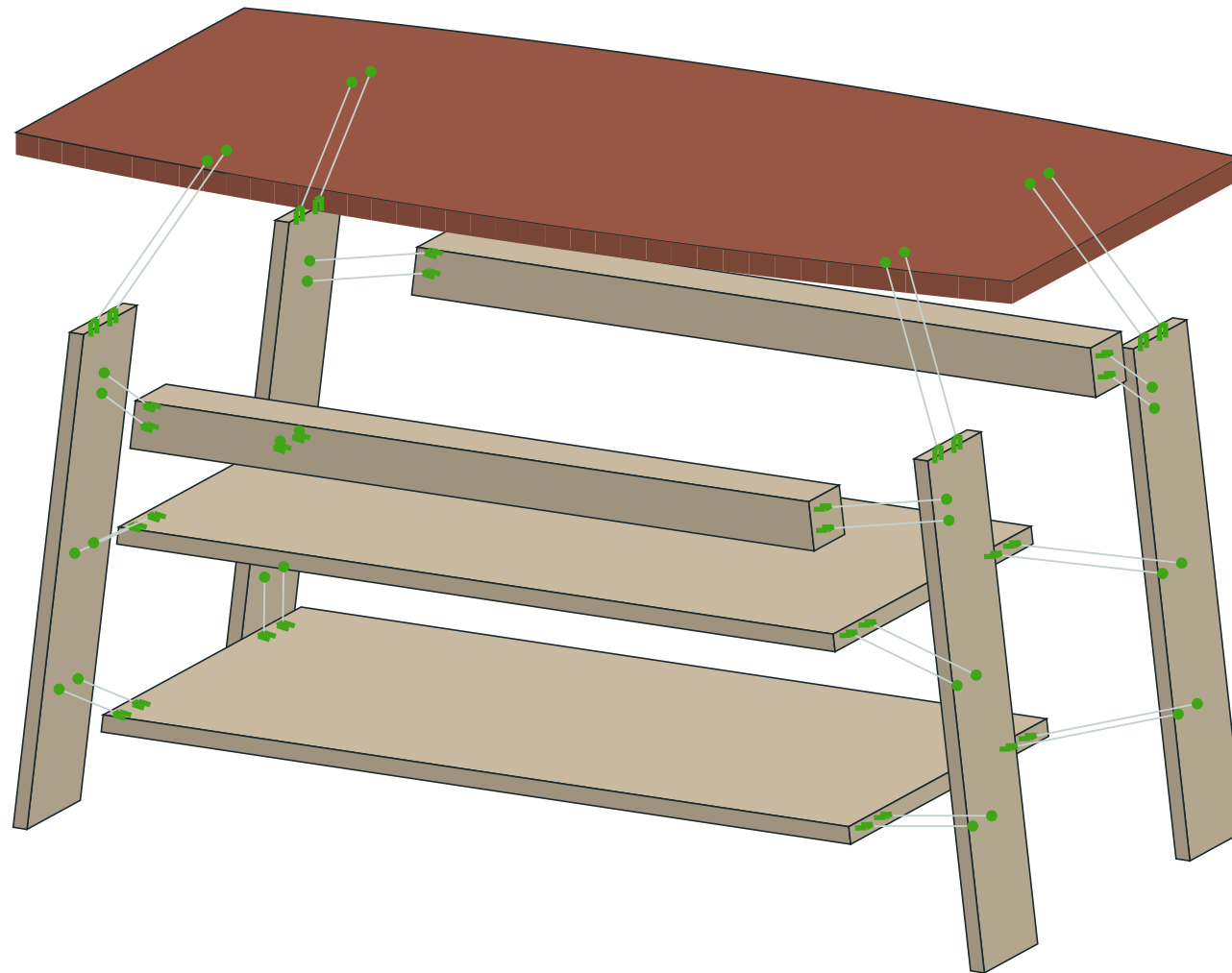
Green U-shaped fittings belong to their carriers. Green dots mark matching receiver anchors. Separation is illustrative.

P09 / BOAT TOP

Receives eight locks on its underside. The four leg tops remain horizontal.

P02 + P03 / SHELVES

Each shelf carries eight locks. Its end bevels match the 8-degree inner leg faces.



32 PHYSICAL LOCKS = 64 CARRIER BORES + 32 RECEIVER ROUTES

Shelf joints: 16 locks. Front and rear apron joints: 8 locks. Leg-to-top joints: 8 locks. The apron rails have their own fittings; no support part is left unattached. All fitting anchors and orientations are listed on sheets 5-6.

Component schedule / exact envelope

World axes: X left/right, Y up, Z front/back. Front is negative Z. All dimensions are millimeters.

ID	COMPONENT	ENVELOPE X x Y x Z	CENTER X, Y, Z	PROFILE
P01	Front-left splayed leg	127.313728, 728, 140	-663.656864, 364, -190	Sheet 4
P02	Lower bevel shelf	1354.032755, 25, 520	0, 192.5, 0	Sheet 4
P03	Upper bevel shelf	1297.816421, 25, 520	0, 392.5, 0	Sheet 4
P04	Front bevel apron	1235.978454, 70, 80	0, 635, -190	Sheet 4
P05	Rear-left splayed leg	127.313728, 728, 140	-663.656864, 364, 190	Sheet 4
P06	Rear bevel apron	1235.978454, 70, 80	0, 635, 190	Sheet 4
P07	Front-right splayed leg	127.313728, 728, 140	663.656864, 364, -190	Sheet 4
P08	Rear-right splayed leg	127.313728, 728, 140	663.656864, 364, 190	Sheet 4
P09	Boat-shaped top	1800, 32, 700	0, 744, 0	Curved top

GEOMETRY RULES

Inner leg half-gap: $e(Y) = 600 + (728 - Y) \times \tan(8 \text{ degrees})$. Left inner face is $X=-e(Y)$; right inner face is $X=+e(Y)$. Outer faces are 25 mm farther out horizontally. The leg feet and top ends are horizontal, so these are bevel-cut profiles, not rotated rectangular boxes.

Shelves span $Y=180..205$ and $Y=380..405$. Aprons span $Y=600..670$. Every beveled end follows $X=\pm e(Y)$. Legs are 140 deep, centered at $Z=-190$ and $+190$. Shelves are 520 deep. Aprons are 80 deep, centered at $Z=-190$ and $+190$.

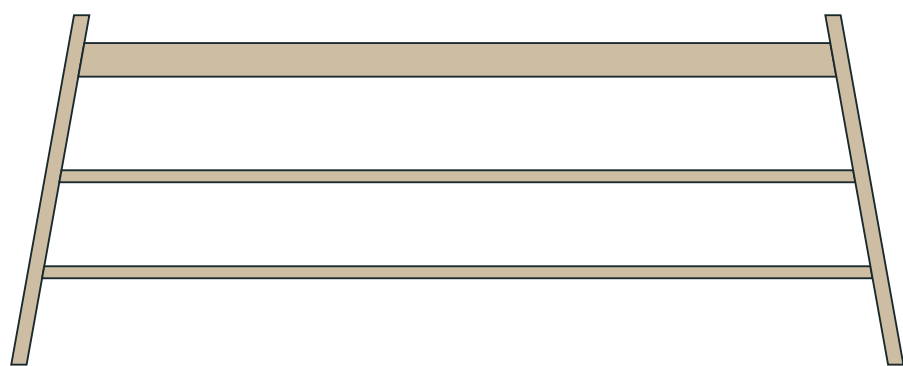
Top outline: $Z=\pm[350 - 50(X/900)^2]$, $-900 \leq X \leq 900$. Width is 700 at the center and 600 at the ends. Underside $Y=728$; finished top $Y=760$. Support color: #c8b89c. Top color: #9b5945. Envelope figures are rounded to six decimals; the formulas define the geometry.

The profiles / true bevel geometry

World (X, Y) vertices in perimeter order. Extrude each closed profile through its stated Z range.

ID	PROFILE	WORLD VERTICES (X, Y)	Z RANGE
P01	Splayed leg	(-727.313728, 0) (-702.313728, 0) (-600, 728) (-625, 728)	-260, -120
P02	Beveled shelf	(-677.016377, 180) (677.016377, 180) (673.502857, 205) (-673.502857, 205)	-260, 260
P03	Beveled shelf	(-648.90821, 380) (648.90821, 380) (645.39469, 405) (-645.39469, 405)	-260, 260
P04	Beveled apron	(-617.989227, 600) (617.989227, 600) (608.151368, 670) (-608.151368, 670)	-230, -150
P05	Splayed leg	(-727.313728, 0) (-702.313728, 0) (-600, 728) (-625, 728)	120, 260
P06	Beveled apron	(-617.989227, 600) (617.989227, 600) (608.151368, 670) (-608.151368, 670)	150, 230
P07	Splayed leg	(702.313728, 0) (727.313728, 0) (625, 728) (600, 728)	-260, -120
P08	Splayed leg	(702.313728, 0) (727.313728, 0) (625, 728) (600, 728)	120, 260

FRONT PROFILE / TRUE 8-DEGREE SPLAY



The shelf and apron ends use the same plane as the leg faces. Each lock approaches normal to that sloping plane and then slides tangentially along its receiver route. A straight box approximation would leave gaps or misplaced anchors.

Use the vertex profiles and formulas together. Do not replace the bevel with a rotated shelf or move connector anchors to the center of the overall bounding box.

Connector schedule / L01-L16

Each row is one complete lock. C is the outward carrier normal; U is the longitudinal slot axis.

LOCK	CARRIER	RECEIVER	FINAL ANCHOR P (X, Y, Z)	NORMAL C (X, Y, Z)	SLOT U (X, Y, Z)
L01	P02	P01	-675.259617, 192.5, -215	-0.990268, 0.139173, 0	0, 0, -1
L02	P02	P01	-675.259617, 192.5, -165	-0.990268, 0.139173, 0	0, 0, -1
L03	P02	P05	-675.259617, 192.5, 165	-0.990268, 0.139173, 0	0, 0, 1
L04	P02	P05	-675.259617, 192.5, 215	-0.990268, 0.139173, 0	0, 0, 1
L05	P02	P07	675.259617, 192.5, -215	0.990268, 0.139173, 0	0, 0, 1
L06	P02	P07	675.259617, 192.5, -165	0.990268, 0.139173, 0	0, 0, 1
L07	P02	P08	675.259617, 192.5, 165	0.990268, 0.139173, 0	0, 0, 1
L08	P02	P08	675.259617, 192.5, 215	0.990268, 0.139173, 0	0, 0, 1
L09	P03	P01	-647.15145, 392.5, -215	-0.990268, 0.139173, 0	0, 0, -1
L10	P03	P01	-647.15145, 392.5, -165	-0.990268, 0.139173, 0	0, 0, -1
L11	P03	P05	-647.15145, 392.5, 165	-0.990268, 0.139173, 0	0, 0, 1
L12	P03	P05	-647.15145, 392.5, 215	-0.990268, 0.139173, 0	0, 0, 1
L13	P03	P07	647.15145, 392.5, -215	0.990268, 0.139173, 0	0, 0, 1
L14	P03	P07	647.15145, 392.5, -165	0.990268, 0.139173, 0	0, 0, 1
L15	P03	P08	647.15145, 392.5, 165	0.990268, 0.139173, 0	0, 0, 1
L16	P03	P08	647.15145, 392.5, 215	0.990268, 0.139173, 0	0, 0, 1

KEEP THE TWO POSTS IN THE CARRIER

Carrier bore entry centers: P - 8U and P + 8U; diameter 5, depth 16 inward along -C. Concept route: P - 9U to P + 23U, with wide entry toward +U; cut into the receiver along +C. The sloped normals are intentional. They keep the fitting aligned with the bevel instead of pointing horizontally through the wrong face.

Connector schedule / L17-L32

Each row is one complete lock. C is the outward carrier normal; U is the longitudinal slot axis.

LOCK	CARRIER	RECEIVER	FINAL ANCHOR P (X, Y, Z)	NORMAL C (X, Y, Z)	SLOT U (X, Y, Z)
L17	P04	P01	-615.17841, 620, -190	-0.990268, 0.139173, 0	0, 0, -1
L18	P04	P01	-610.962185, 650, -190	-0.990268, 0.139173, 0	0, 0, -1
L19	P04	P07	615.17841, 620, -190	0.990268, 0.139173, 0	0, 0, 1
L20	P04	P07	610.962185, 650, -190	0.990268, 0.139173, 0	0, 0, 1
L21	P06	P05	-615.17841, 620, 190	-0.990268, 0.139173, 0	0, 0, -1
L22	P06	P05	-610.962185, 650, 190	-0.990268, 0.139173, 0	0, 0, -1
L23	P06	P08	615.17841, 620, 190	0.990268, 0.139173, 0	0, 0, 1
L24	P06	P08	610.962185, 650, 190	0.990268, 0.139173, 0	0, 0, 1
L25	P01	P09	-612.5, 728, -215	0, 1, 0	0, 0, 1
L26	P01	P09	-612.5, 728, -165	0, 1, 0	0, 0, 1
L27	P05	P09	-612.5, 728, 165	0, 1, 0	0, 0, 1
L28	P05	P09	-612.5, 728, 215	0, 1, 0	0, 0, 1
L29	P07	P09	612.5, 728, -215	0, 1, 0	0, 0, 1
L30	P07	P09	612.5, 728, -165	0, 1, 0	0, 0, 1
L31	P08	P09	612.5, 728, 165	0, 1, 0	0, 0, 1
L32	P08	P09	612.5, 728, 215	0, 1, 0	0, 0, 1

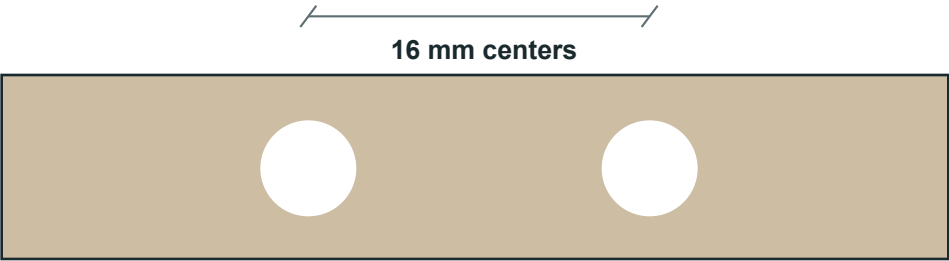
KEEP THE TWO POSTS IN THE CARRIER

Carrier bore entry centers: P - 8U and P + 8U; diameter 5, depth 16 inward along -C. Concept route: P - 9U to P + 23U, with wide entry toward +U; cut into the receiver along +C. The sloped normals are intentional. They keep the fitting aligned with the bevel instead of pointing horizontally through the wrong face.

Inside the Lockdowel connection

Detail diagrams are schematic. Dimensions marked "concept" define the film, not production tooling.

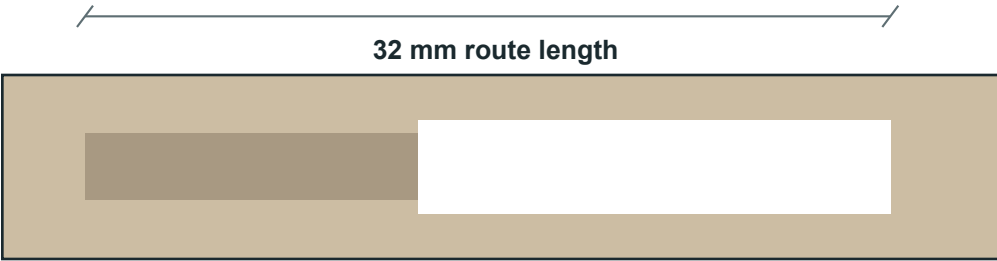
A / TWO BORES IN THE CARRIER



2 x diameter 5 mm, 16 mm deep

Posts point inward; the bridge remains outside.

B / ROUTE IN THE RECEIVER

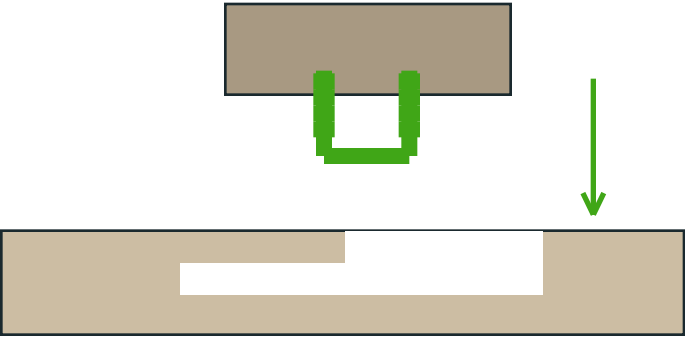


Narrow capture region

Wide entry region

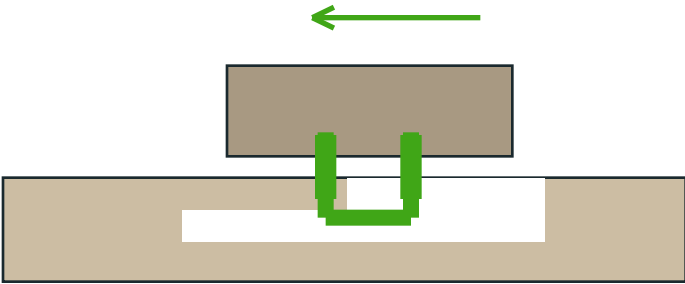
Shallow depth 3; deep depth 6 to 6.3 mm. See sheet 9.

C / ENTER FIRST. SLIDE SECOND.



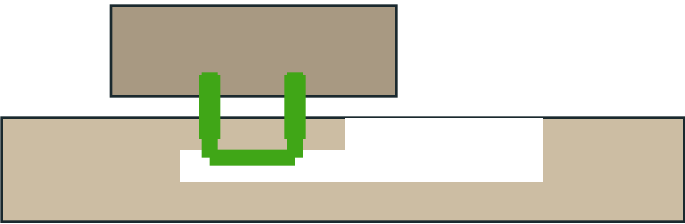
1 / APPROACH

Move perpendicular to the receiver. The fitting travels with its carrier.



2 / BRIDGE ENTERED

The bridge is in the wide opening. Now move along the slot, not into it again.



3 / SLID UNDER THE LIP

The retaining shoulder captures the bridge. The two posts remain in the carrier.

CONCEPT PARAMETERS: 21 mm bridge length; 6.35 mm undercut width; 35 mm approach clearance; 14 mm locking travel. Exact SKU cross-section and machining tolerances need engineering confirmation. Use the original routing sheet for source dimensions.

Sixteen movements / a complete locking film

Start with P01 supported upright. All 32 U-shaped fittings are preinstalled in their designated carriers.

STEP	MOVING COMPONENT	ACTION / mm	FROM CENTER (X, Y, Z)	TO CENTER (X, Y, Z)
01	P02 / Lower bevel shelf	Enter 35	34.659382, 187.628941, -14	0, 192.5, -14
02	P02 / Lower bevel shelf	Slide 14	0, 192.5, -14	0, 192.5, 0
03	P03 / Upper bevel shelf	Enter 35	34.659382, 387.628941, -14	0, 392.5, -14
04	P03 / Upper bevel shelf	Slide 14	0, 392.5, -14	0, 392.5, 0
05	P05 / Rear-left splayed leg	Enter 35	-698.316246, 368.871059, 176	-663.656864, 364, 176
06	P05 / Rear-left splayed leg	Slide 14	-663.656864, 364, 176	-663.656864, 364, 190
07	P04 / Front bevel apron	Enter 35	34.659382, 630.128941, -204	0, 635, -204
08	P04 / Front bevel apron	Slide 14	0, 635, -204	0, 635, -190
09	P07 / Front-right splayed leg	Enter 35	698.316246, 368.871059, -204	663.656864, 364, -204
10	P07 / Front-right splayed leg	Slide 14	663.656864, 364, -204	663.656864, 364, -190
11	P06 / Rear bevel apron	Enter 35	34.659382, 630.128941, 176	0, 635, 176
12	P06 / Rear bevel apron	Slide 14	0, 635, 176	0, 635, 190
13	P08 / Rear-right splayed leg	Enter 35	698.316246, 368.871059, 176	663.656864, 364, 176
14	P08 / Rear-right splayed leg	Slide 14	663.656864, 364, 176	663.656864, 364, 190
15	P09 / Boat-shaped top	Enter 35	0, 779, -14	0, 744, -14
16	P09 / Boat-shaped top	Slide 14	0, 744, -14	0, 744, 0

SHOWCASE THE BEVEL, THEN THE FOUR-LEG TOP CONNECTION

Use a fixed assembly view, with the close-up showing how the angled joint accepts the bridge before the slide. Do not repeat the entry movement in place of locking. Finish with the curved top lowering and sliding onto all four leg tops, then hold on the complete nine-part table.

Dimensioned concept only. The profile and sequence have been locally checked, but exact connector SKU surfaces, machining tolerances and structural loading require engineering confirmation.

Original Mini routing reference

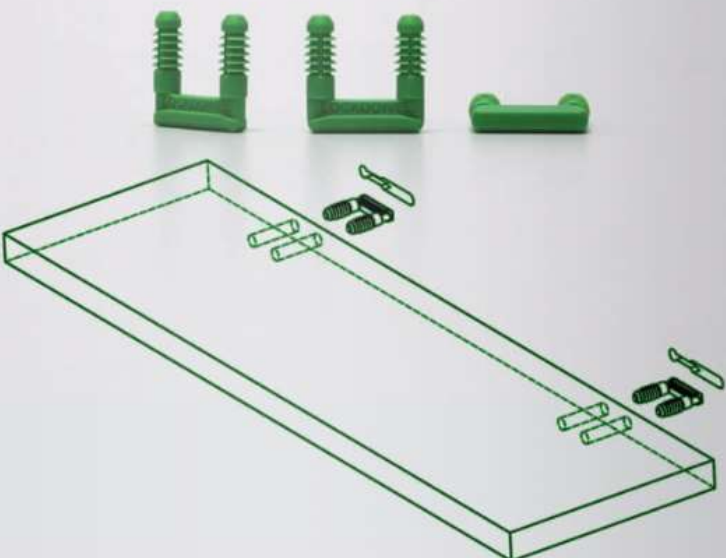
Source: supplied "LD 5 inch E900BP-mini drawer box details (1).pdf", page 11. Reference hardware only; no drawer is included.

E900BPMini Mini Nylon Barbed Channel Lock

Specification

- Removable Sturdy Joint
- Pull Force: 70+ lbs
- Sheer Force: Over 50 lbs
- Requires CNC routing capability and edge boring machine
- Best used in Particle Board, Plywood, MDF, and selected plastic materials (Min. Thickness: 1/2")

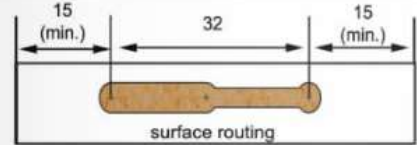
- Drilling: Ø 5 mm and 16 mm spacing
- Material: Nylon 6-6



10

Patent Pending

Mini Barbed Channel Lock Routing Instructions



surface routing



Edge boring for barbed post installation

Diameter: 5*
Depth: 16



surface routing: section view

Plunge Cut: Z: -3.0

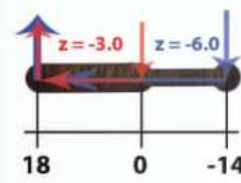
Slot Cut: X: 0 to 18

Plunge Cut: Z: -6.0

Key Slot: X:-14 to 18

(Allowable Depth Range is -6.0 to -6.3)

Route RED first. Then BLUE.



18 0 -14

• All measurements are in mm.

• If using MDF, replace 5mm Drill Bit with 5.2mm Drill Bit.

11

Patent Pending